

Dashboard Project Report

Akash Prasad | [LinkedIn](#)

Title: Sales & Profit Dashboard - (Sample Superstore) – (MS Power Bi)

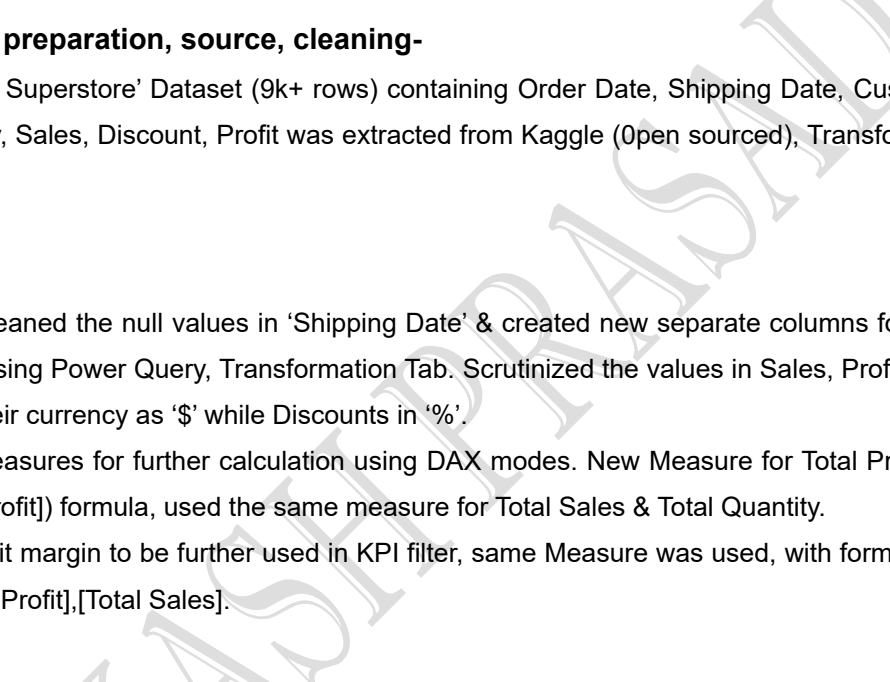
1. Statement of Problem

The traditional methods of reporting statistics relied heavily on spreadsheet based or formula based intervention, which requires a particular research background for one to understand datasets. Here, dataset Global Superstore was used to identify the growth of sales & profit (in %) in different regions of US. The goal was to create an easy understandable visual & interactive dashboard showing charts, bar graphs, and using measure values.

2. Data sourcing & preparation, source, cleaning-

Source- The 'Sample Superstore' Dataset (9k+ rows) containing Order Date, Shipping Date, Customer ID, State, City, Region, Sub-Category, Sales, Discount, Profit was extracted from Kaggle (Open sourced), Transformed in Power Query & visualized in Bi tool.

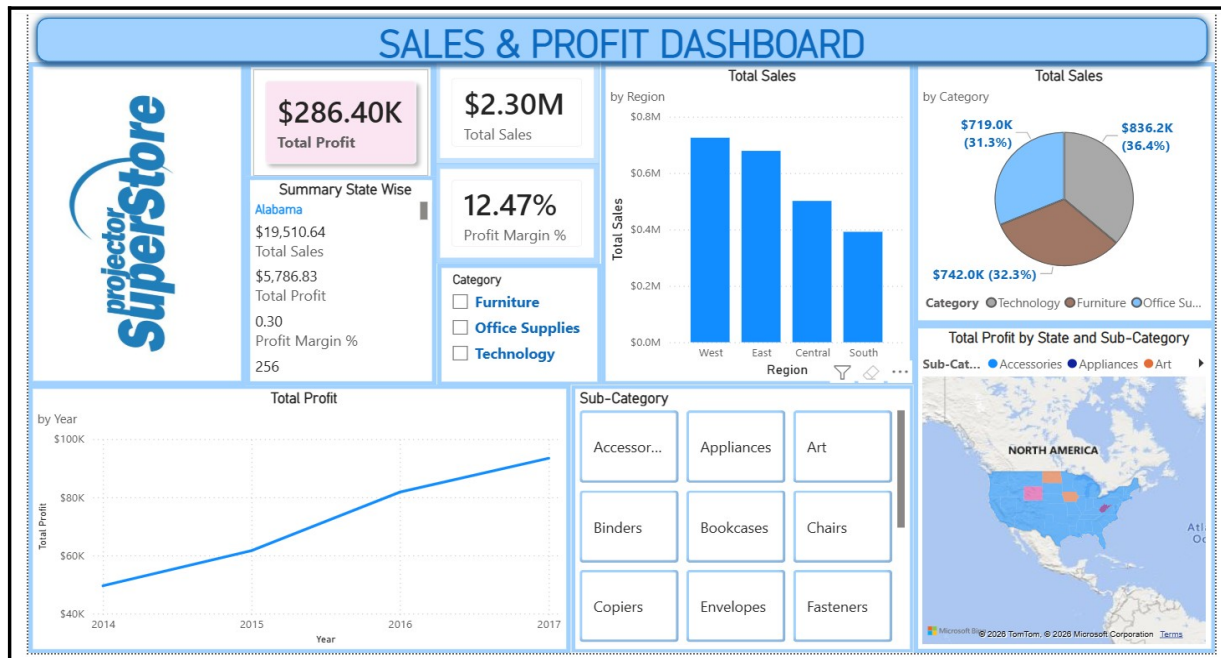
Data Cleaning -

- Checked & cleaned the null values in 'Shipping Date' & created new separate columns for Year, Month & Start of Month by using Power Query, Transformation Tab. Scrutinized the values in Sales, Profit as decimal numbers & changed their currency as '\$' while Discounts in '%'.

- Used New Measures for further calculation using DAX modes. New Measure for Total Profit as calculated with SUM(name[Profit]) formula, used the same measure for Total Sales & Total Quantity.
- To derive Profit margin to be further used in KPI filter, same Measure was used, with formula, Profit Margin % = DIVIDE([Total Profit],[Total Sales]).

3. Methodology

1. Created an interactive KPI card for newly created measured column for dashboard 'Total Profit' from Visualization tab, added effects, colored borders for highlighted view.
2. A state wise summary was added, interactive- scroll friendly to easily view the Sales, Profit and Margins.
3. Category was dragged from Data Tab to Visualization>Slicer in order to show three categories that the store shipped & a similar interactive slicer filter was created for Sub-category to quickly identify profits or sales from one particular shipped item.
4. Total Profit by State & Sub-Category was included in a MAP, for a comparative overview of Profits from particular accessories in that region, the region is highlight with different colors for different sub-category.
5. Region wise Bar chart was made to compare and find out maximum Total Sales in millions for a region, where as Line chart for Total Profit by Year.
6. Pie chart indicating Total Sales by Category, colored, differentiated & labeled with legends in both Numerical/\$ & %.

Visualization



4. Key insights & Recommendations

- Macro level key insights can be checked, that keeps the business informed about their volume of Sales that is in Millions, against their efficiency in total profit and profit margins. A region & category wise filter enabled high-sales period & regions, it showed in which state the profitability of particular category of accessory is low.
- Yearly trajectory of Line chart showing an upward trend for Total Profits, KPI indicating positive numbers & Pie chart showing %-wise Sales, indicates there is no heavy discount affecting at country level.
- With in interactive map linked with state summary, it is possible to bridge the gap between higher sales and higher discounts in a geographic variation.
- Further, the dataset calculated with DAX modes, can be used to run a linear regression or time series analysis to find out regional variations of different states.
- This dataset can be put into use in other analytical tools like python where numpy & pandas can run prediction models & matplotlib to create plots or charts to display the regression lines. That can identify the lowest sales, higher discount and less profit margin of a firm in long term.

Dataset Source: Kaggle – Sample Superstore' Dataset